

3 A a-t graph obtained from gradient of v-t graph

4 B Taking downwards as positive,
 $S = ut + \frac{1}{2} at^2$
 $= -3.00 \times 5.00 + \frac{1}{2} 9.81(5.00)^2 = 108 \text{ m}$

5 C By conservation of momentum: